

EX. NO: 4

Simple Knowledge-Based AI System

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Aim:

To develop a simple knowledge-based AI system in Python that diagnoses possible illnesses based on user-reported symptoms using predefined rules.

Procedure:

- 1. Design the knowledge base:**

Define a set of rules that associate specific symptom combinations with possible diagnoses.

- 2. Collect user input:**

Prompt the user to answer a series of yes/no questions regarding common symptoms.

- 3. Store symptoms:**

Record symptoms for which the user answers "yes".

- 4. Apply rules to diagnose:**

Match the collected symptoms against the knowledge base to infer a diagnosis.

- 5. Display the diagnosis:**

Show the inferred condition or advise consulting a doctor if symptoms do not match known patterns.

Program Coding:

```
def diagnose(symptoms):
```

```
    # Knowledge base: rules
```

```
    if "fever" in symptoms and "cough" in symptoms and "fatigue" in symptoms:
```

```
        return "You might have the flu."
```

```
    elif "headache" in symptoms and "nausea" in symptoms:
```

```
        return "You might have a migraine."
```

```

elif "sore throat" in symptoms and "fever" in symptoms:

    return "You might have a throat infection."

else:

    return "Diagnosis unclear. Please consult a doctor."
# Get input from user
print("Welcome to the Simple Medical Diagnosis System.")
print("Please answer the following questions with 'yes' or 'no'.")
# Collect user symptoms
symptoms = []
questions = {

    "fever": "Do you have a fever?",
    "cough": "Do you have a cough?",
    "fatigue": "Are you feeling tired or fatigued?",
    "headache": "Do you have a headache?",
    "nausea": "Are you feeling nauseous?",
    "sore throat": "Do you have a sore throat?"
}

for symptom, question in questions.items():
    answer = input(question + " ").strip().lower()
    if answer == "yes":
        symptoms.append(symptom)
# Run diagnosis
result = diagnose(symptoms)
print("\nDiagnosis:", result)

```

Output:

```

Welcome to the Simple Medical Diagnosis System.
Please answer the following questions with 'yes' or 'no'.
Do you have a fever? yes
Do you have a cough? yes
Do you have a headache? no
Do you have a sore throat? no
Do you have nausea? no

```

Do you have fatigue? yes
Diagnosis: You might have the flu.

Result:

Thus, the simple knowledge-based AI system was successfully implemented.